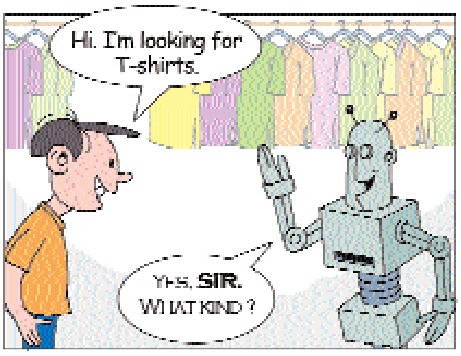
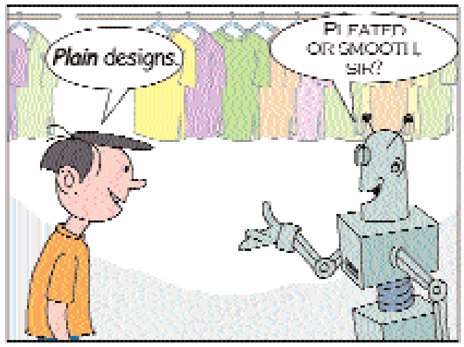
Bot with an understanding

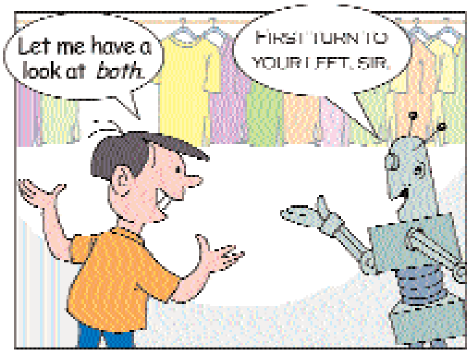
An advanced NLU could eliminate the need for everybody in a clothes shop, except perhaps the guy at the counter. Here's a typical scene at a regular clothes shop, only, it has NLU-capable bots. Notice the words in italics and what they could have meant.



The bot figured the person's gender



'*No-frills designs*', not 'T-shirts with aeroplanes on them'



'*Both*' meant both kinds of *shirts*

the computer, to some extent, understands the text it is speaking.

Why NLP is hard

NLP is probably the best example of what AI is all about. But there are quite a few hiccups on the road to building the perfect NLUs.

Suppose one was to say, "Garfield's sitting back in his chair and relaxing". You would definitely understand it to mean that Garfield's relaxation is, at least partly, a result of his sitting in his chair. But a naïve NLU system would take it to mean that Garfield is doing two activities—(a) sitting on his chair, and (b) relaxing; the NLU doesn't co-relate the two individual bits of information to comprehend the sentence in its entirety. An NLU system also needs to fill in appropriate pieces to an incomplete sentence such as, 'It's hot today.' A naïve NLU would wonder what is hot today, while an evolved one would naturally conclude that you mean the weather.

A smattering of common sense also goes a long way. The sentence, 'It's late and I want to go' tells you that I want to get out from the place I am in now. But, it's difficult to make a machine understand that I want to go 'out', without explicitly using that word.

While using slang, parts of speech are often used interchangeably. A noun is sometimes used as a verb, and vice-versa. So if you say, "House him somewhere", an NLU system might generate an error, saying that 'house' is a noun. A trivial solution to this problem would be, "just supply all this information to the program". But that amount of information is huge, and the task of using it correctly, humongous.

References to other words, is a classic example of what makes a NLP system's work hard. Suppose you say, "Garfield and Jon are funny. I like the way he gets disgusted when the cat pours mustard in his coffee." Now look at the 'he' and the 'his' in that second sentence. It's easy for us to figure out that they refer to Jon, but for a machine, even if you tell it that Garfield is the name of the cat, it will not automatically know who the 'he' and 'his' pertain to.

The need for phonetic information is extremely essential in speech-recognition systems, since a word can be identified based only on how it sounds. There needs to be a knowledge base that links words and sounds. The problem is that, when words are spoken at a normal pace, the sound is not the same as when they are spoken as distinct words.

Often a word can mean different things, in different contexts. For example, if the setting is an infotech office party, then 'host' would probably mean a server, and not the person who is hosting the party.

The logic-and rule-based approach

There are a few fundamental approaches to NLP, including statistical approaches. Essentially, the three phases in bringing a textual sentence to the desired form are the syntactic (structure), semantic (meaning) and the final representation.

The syntactic phase takes care of the parsing of the sentence, i.e., the breaking down of the sentence into its component parts of speech such as nouns, verbs and adjectives. Parsing for NLP has a few significant differences from programming language parsing. Among them is the fact that there is no standard grammar to work with. Writing a grammar for a natural language is not easy. In processing natural language, part of speech organisation is essential, as opposed to word-by-word parsing

Research Groups in India

The NLP focus of the AU-KBC Research Centre of Anna University, Chennai, is Indian languages. The group works on building lexical resources such as dictionaries, a Tamil version of WordNet (an online lexical reference system) and tagged corpora (text that has been tagged in a desired manner). These are essential for researchers working on various areas of NLP. The group built the Tamil-Hindi Machine Aided Translation system, which has achieved accuracy in the range of 75 per cent. They also have the Tamil Morphological Analyzer that can handle nearly 3.5 million word forms, including compound words, with accuracy as high as 95 per cent!

A group at IIT Kanpur is also working on projects such as Anglabharti—the Machine Aided Translation system from English to Indian Languages, as well as Speech-to-Speech Translation, Lexical Knowledge-Base Development, Transliteration and other linguistic projects.